Cell

Charting a Course to a Cure

In this issue of *Cell* and in papers appearing in *Cell Reports*, *Cancer Cell*, *Cell Systems*, and *Immunity*, we are proud to present the Pan-Cancer Atlas. The Pan-Cancer Atlas is the culmination of more than a decade of work from The Cancer Genome Atlas (TCGA) consortium. By analyzing over 11,000 tumors from 33 of the most prevalent forms of cancer (see SnapShot, page 530), the atlas provides a uniquely comprehensive, in-depth, and cohesive understanding of how, where, and why tumors arise in humans. TCGA has long been at the forefront of cancer-genome sequencing, but the effort also incorporates analysis of cancer transcriptomes, proteomes, methylomes, clinical data, and imaging. Through its integration of multiple data types and its ability to be mined for actionable information, the Pan-Cancer Atlas can be likened to a "Google Earth" for human cancer research.

To open the collection, Carolyn Hutter and Jean Claude Zenklusen (page 283) provide an overview of the legacy of TCGA and the genesis of the Pan-Cancer Atlas project. The atlas consists of 27 papers divided into three categories: cell-of-origin patterns, oncogenic processes, and oncogenic pathways. Each category is anchored by a "marker" paper, appearing in *Cell* and providing a summary of the core findings for that topic. Multiple companion papers within these three categories explore individual subtopics in depth and are distributed among multiple Cell Press journals. To aid in navigating the expansive terrain the collection covers, we have created a dedicated web portal for readers and researchers (www.cell.com/consortium/pancanceratlas). Additionally, a webinar featuring project leaders Katherine Hoadley, Nikolaus Schultz, and Li Ding will provide a guided tour through some of the altas's insights (http://www.cell.com/webinars).

Among its numerous highlights, the Pan-Cancer Atlas reveals a new clustering of tumor types that will inform future clinical trials, grouping tumors based on molecular similarity and with a greater understanding of how their tissue of origin influences a cancer's features. The collection also uncovers potential vulnerabilities that will aid in the development of combination therapies and personalized medicine. The atlas sheds new light on the collaboration between germline genetic variants and somatic mutations in cancer progression and the influence of mutation on cell signaling and immune-cell composition, which will help prioritize the development of new treatments and immunotherapies for a wide range of cancers. As a singular point of reference, the Pan-Cancer Atlas will have a long-term impact on setting priorities for new treatment development, combination therapies, and personalized medicine. Its data will be mined far into the future.

The publication of the Pan-Cancer Atlas also signifies a major point of transition: the culmination of the decade-long TCGA effort (see Voices, page 281). To mark this turning point, Cell Press and TCGA are jointly organizing a Cell Symposium from September 27 to 29, 2018, in Washington, D.C. (http://www.cell-symposia.com/tcga-2018/) that will gather prominent speakers (including NIH Director Francis Collins), patient advocates, and other members of the cancer-research community. The meeting is open to all and will highlight the impact of TCGA and multi-omics on cancer research and provide a timely venue for the cancer community to discuss the future of multi-omics efforts.

Publishing a large body of work is an intensive enterprise, and we would like to thank TCGA for their energy and collaborative spirit. Today, as we celebrate the achievements of TCGA, we would also like to take the opportunity at Cell Press to express our commitment and interest in partnering with consortia efforts generally. We are inspired by the collective effort and thinking that drive large-scale scientific collaborations and appreciate the immense value of the data they generate. We know from experience that consortia are diverse in both their make-up and goals and that their needs from a publishing partner are not one-size-fits-all. If you are part of a consortium, we would welcome hearing from you about your ideas, plans, and progress. This is true whether you have a package of papers fully formed and ready to submit, are applying for funding for a multi-lab or institute project, or even if your thoughts are simply percolating at the idea stage. Our goal, for individual papers and for larger collections, is to engage with researchers at all points of the scientific process.

We hope you enjoy the Pan-Cancer Atlas and the other exciting work presented in this issue.

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